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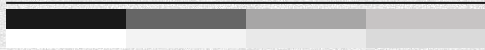
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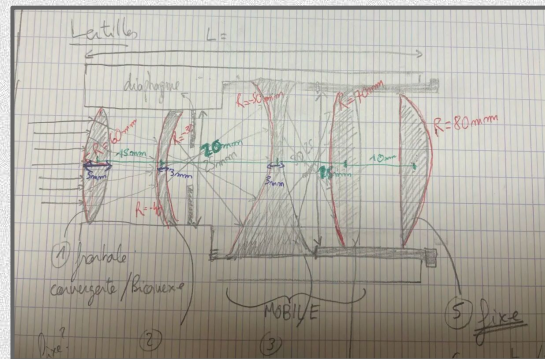
01 OVERALL PROJECT

REAL-WORLD ENGINEERING APPROACH

- Reverse engineering based on two real lenses
(SONY SEL 28-70 & SEL 70-200)
- Dimensions captured via calipers and physical measurements

OPTICAL COMPONENTS

- **Modular architecture:** modeling of interchangeable mounts, accessories...
- **Simplified light flux control:** managing how we converge light onto the sensor



OPTICAL DESIGN LOGIC

- Lens profiles defined using **Descartes' law** for radius of curvature and lens types
- Engineering approximations applied where proprietary manufacturer data was restricted

KNOWLEDGE ADVISOR WORKBENCH

- Checks, warnings, rules, reactions...

28-70MM OPTICAL LENS DESIGN ANALYSIS

Synthesis of Theoretical Calculations, Kinematic Laws, and Industrial Approximations
RAYAN CHENNAOUI - FELIX LAO - LEONARD POUPINET - MATHEO HERVY-CADOUX

1. Optical Foundations (Descartes' Laws)

To model lenses without proprietary manufacturer data, we utilize the Thin Lens Approximation and the Lens Maker's Formula :

$$\Phi = \frac{1}{f'} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Key Variables :

- $n \approx 1.5$ (Standard refractive index for optical-grade polycarbonate/glass)
- R_1, R_2 : Radii of curvature for lens surfaces (mm)
- f' : Focal length of the individual lens element (mm)

2. Engineering Ranges (CAD-Derived Approximations)

Based on physical caliper measurements and internal mechanical constraints :

- **Housing Diameters :**
 - Front Barrel (Large) : $\varnothing \in [60 - 65 \text{ mm}]$
 - Rear Barrel (Small) : $\varnothing \in [50 - 55 \text{ mm}]$
- **Theoretical Curvature Radii (R) :**
 - Front Element (L1) : $R \in [75 - 90 \text{ mm}]$ (Convex)
 - Rear Exit Element (L5) : $R \in [60 - 80 \text{ mm}]$ (Plano-Convex)
- **Lens Thickness (t) :** $t \in [3 - 8 \text{ mm}]$ at the optical center.

3. Chassis Kinematics (Zoom Law)

The displacement of the mobile chassis (Z) is linked to the user-defined focal length (f) via a linear relationship to ensure a smooth transition between 28mm and 70mm.

$$\text{Displacement } Z(f) = a \cdot f + b$$

Prototype Application :

- $f_{min} = 28\text{mm} \rightarrow Z = 0\text{mm}$ (Rear-stop/Wide-angle position)
- $f_{max} = 70\text{mm} \rightarrow Z \approx 22.34\text{mm}$ (Maximum extension/Telephoto)
- **Slope Coefficient :** $a = \frac{22.34}{70-28} \approx 0.531$

4. Knowledge-Based Engineering : Auto-Focus & Safety Checks

The Auto-Focus is simulated by translating a compensator group based on the Subject Distance S :

$$\frac{1}{f'} = \frac{1}{S} + \frac{1}{i} \rightarrow \text{Lens Displacement} = \Delta i$$

Safety & Validation (Knowledge Advisor) :

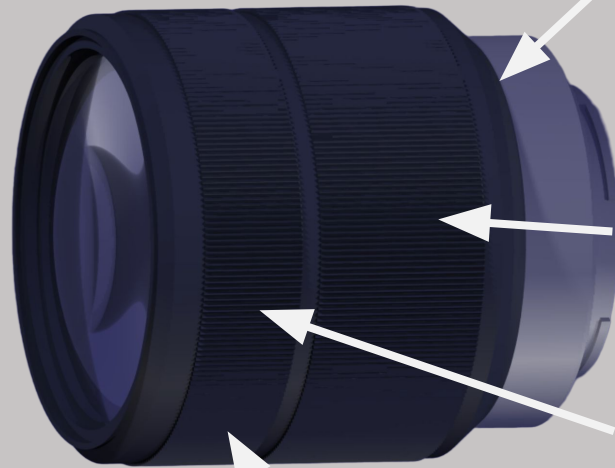
- **Checks :** Automated validation for $[28 - 70\text{mm}]$ and $[90 - 150\text{mm}]$ ranges.
- **Reactions :** Mechanical hard-stop locking and mount compatibility alerts.

Side-by-Side Comparison : SEL 28-70 mm

Real Model



CATIA Modelling

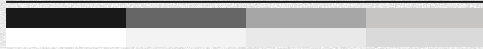


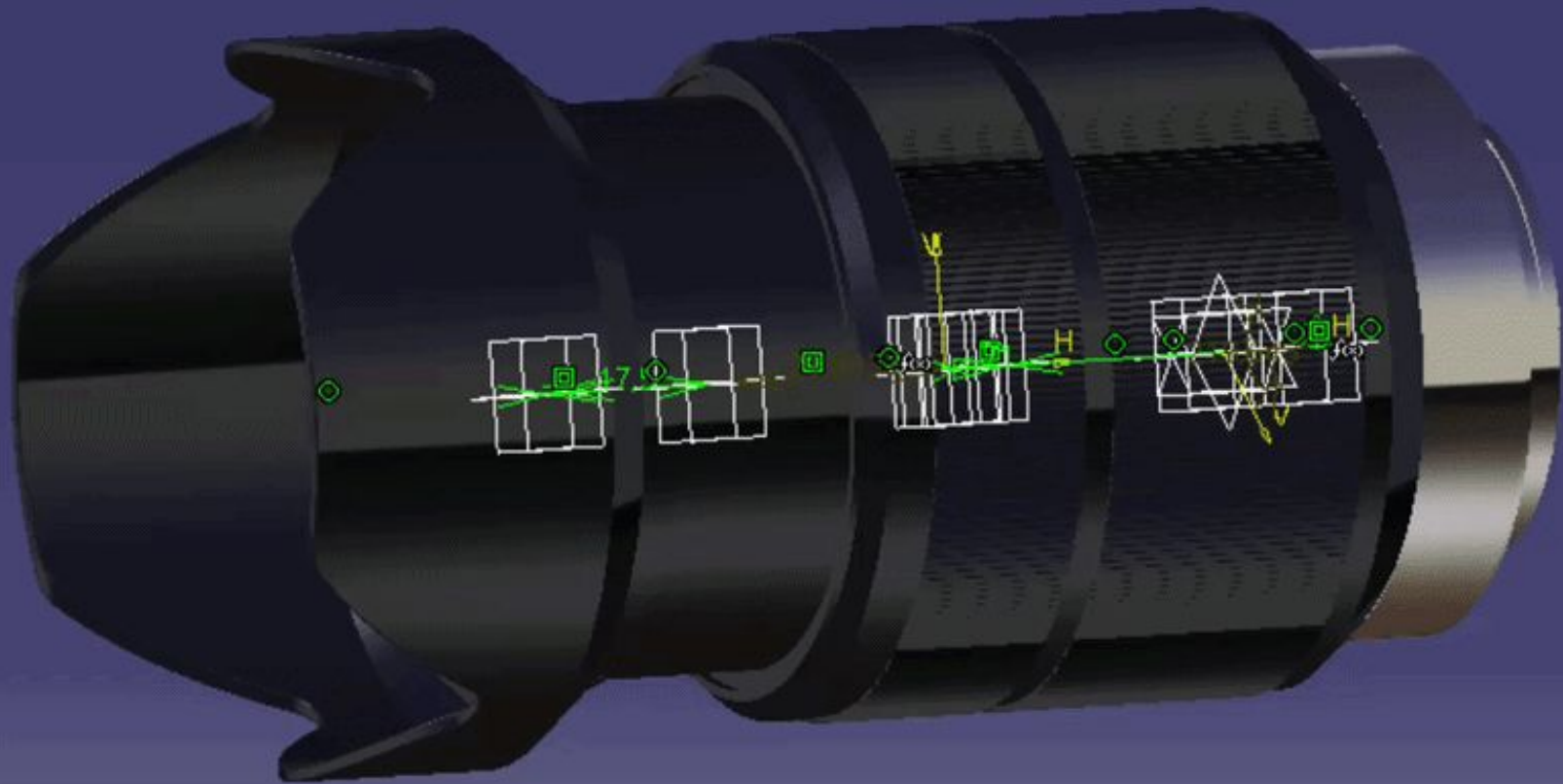
Parameter
for the
Mount

Formula for
the Zoom
Ring

Formula for
the Offset of
the Barrel

Matte Black Finish



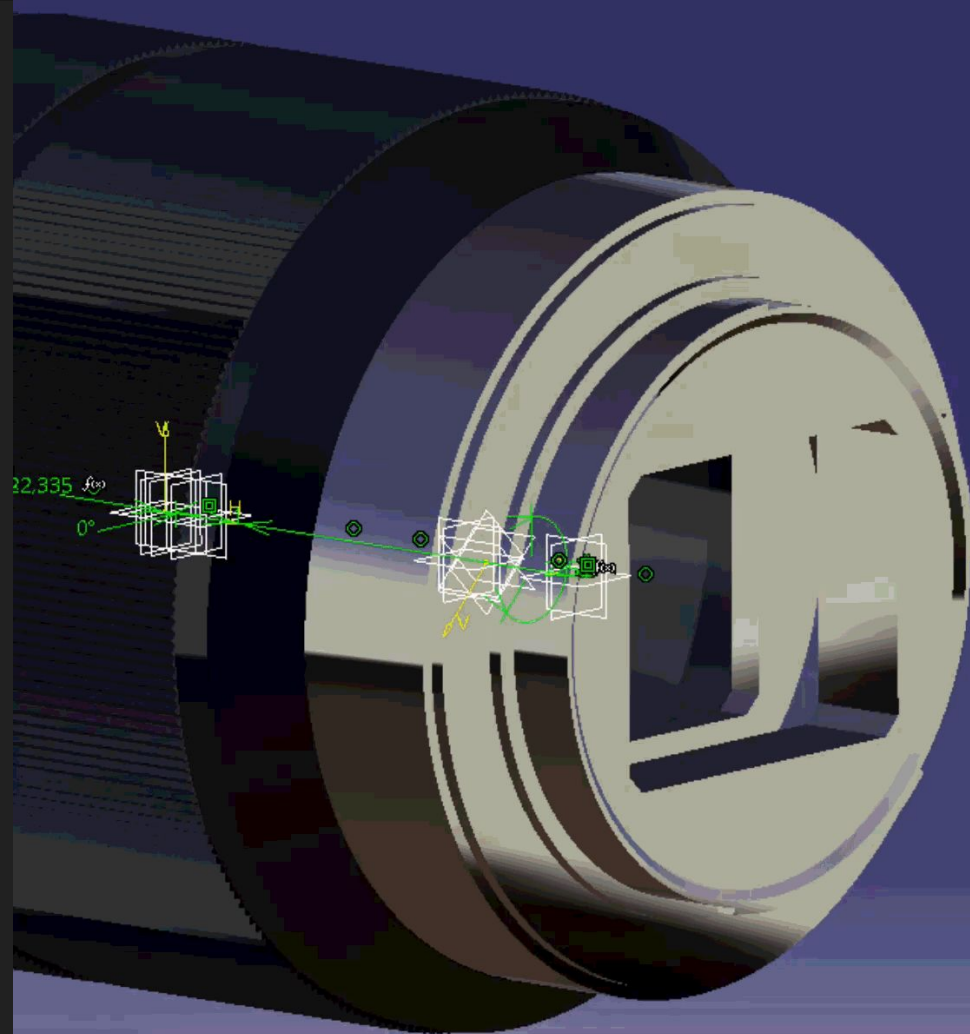


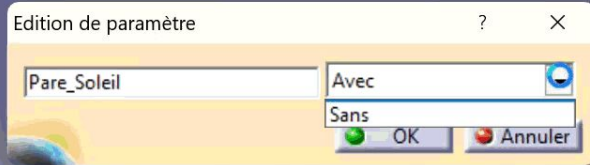
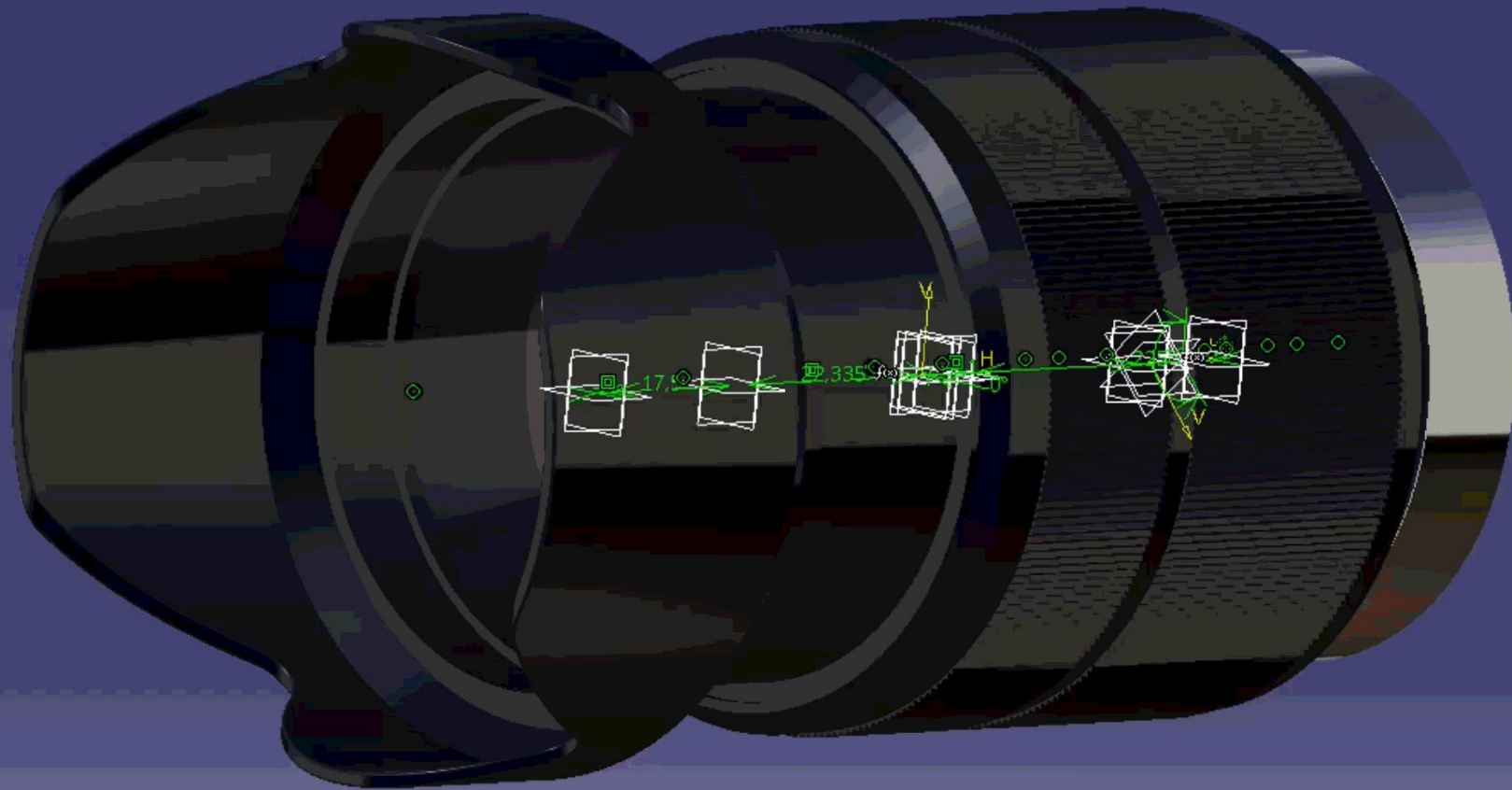
Edition de paramètre ? X

Distance_Focale 69mm

OK Annuler

Mise à jo





Formulas & Constraints

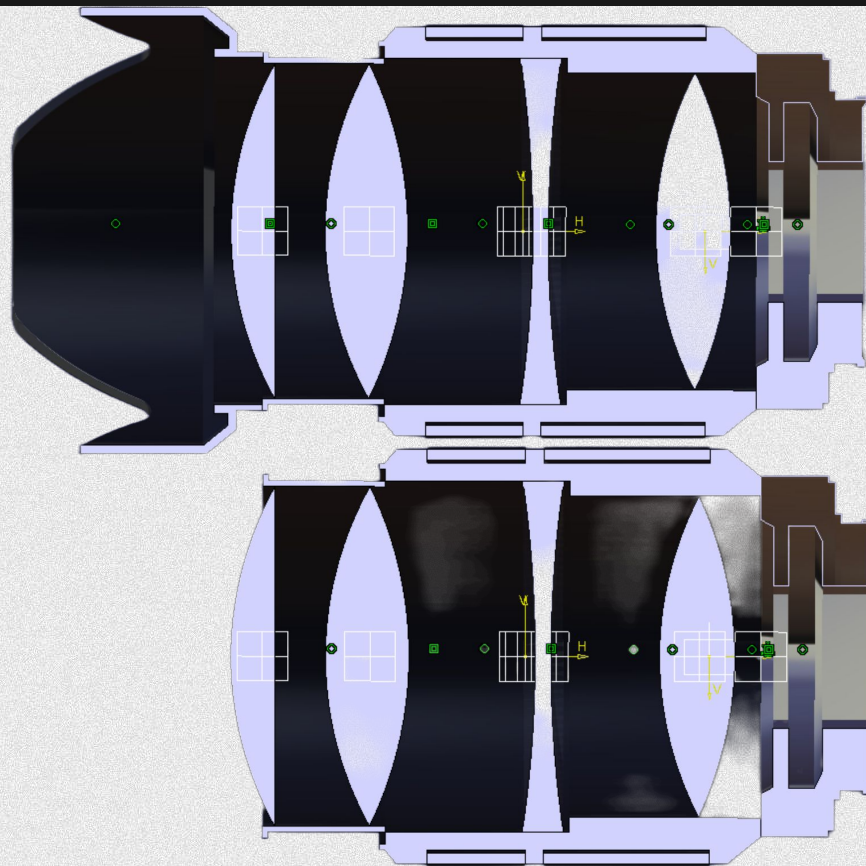
Lens barrel extension (Offset rule)

$$(\text{Distance_Focale} - 28\text{mm}) * (22.335 / 42)$$

Rings: Rotation proportional to focal length

$$\text{Angle} = (((\text{Focal_Dist} / 1\text{mm}) - 28) * (75 / 42)) * 3\text{deg}$$

Options: Lens hood and Mounts (A/E) via String/Boolean parameters.

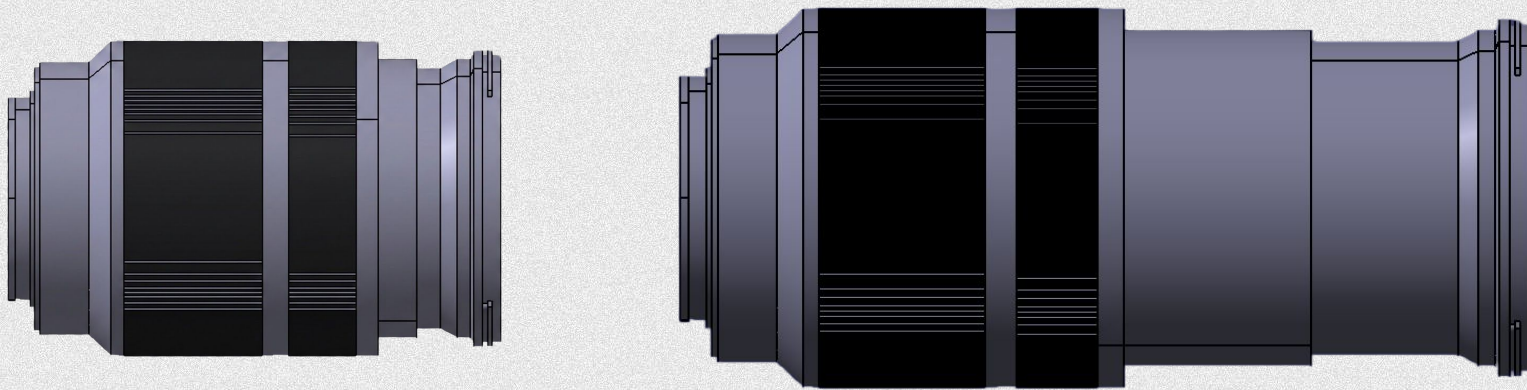


03 Second Lens

Sony 70-200mm

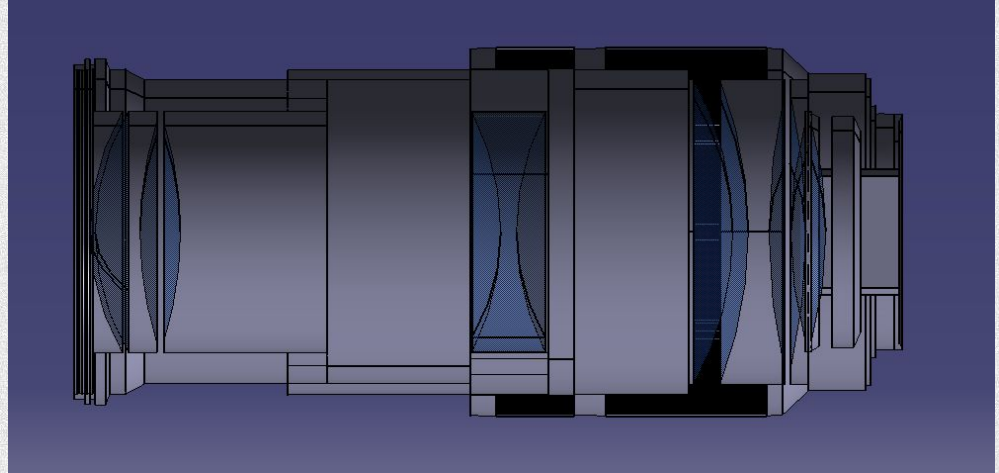


Specification



6 CatPart / Dual-stage telescoping lens

Specifi- cation



2 Moving lens / 1 Free Lens for the Focus / 4 Fix lens

Parameters

2 Rules :

- Moving part
- Petal lens hood

2 checks :

- Min Focal
- Max Focal

```
/*Extension du premier tube ( E1 = 0.2033*F + 54.341)*/  
`Contraintes\Décalage.7\Décalage.7\Offset` = 0.2033mm * `Focal` +  
54.341mm  
/*Rotation Bague 1*/  
`Contraintes\rotationBague1\rotationBague1\Angle` = `Focal` * 1deg
```

```
/*Extension du secnd tube ( E2 = 0.374*F + 12.775)*/  
`Contraintes\Décalage.8\Décalage.8\Offset` = 0.374mm * `Focal` + 12.775mm  
/*Rotation Bague 2*/  
`Contraintes\RotationBague2\RotationBague2\Angle` = (0.374 * `Focal` +  
12.775) * 1deg
```

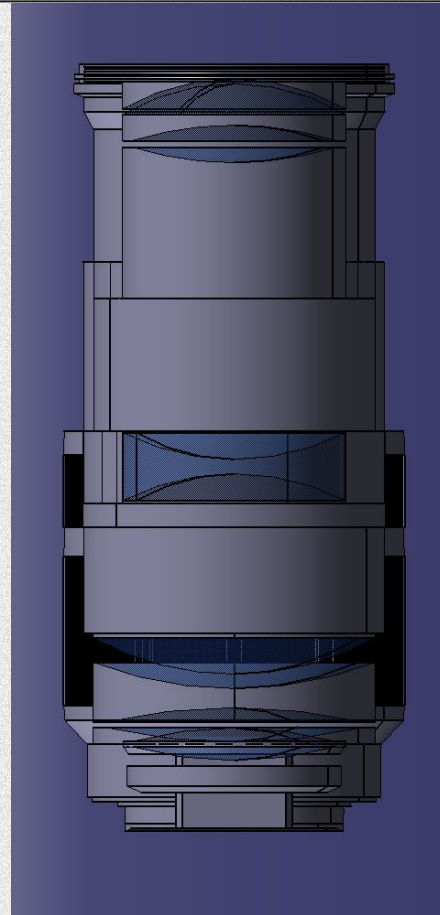
```
/* formule de Descartes : req_p_prime = (f * p) / (p - f)*/  
/* f = focale actuelle, p = distance sujet, req_p_prime = offset au  
capteur*/  
`Contraintes\Décalage.11\Décalage.11\Offset` = ( (0.374mm * `Focal` +  
12.775mm) - (((`Focal` * 1mm) * (`Distance Sujet` * 1000mm)) / ((`Distance  
Sujet` * 1000mm) - (`Focal` * 1mm))))+50mm
```



04 DIFFICULTIES & CONCLUSIONS

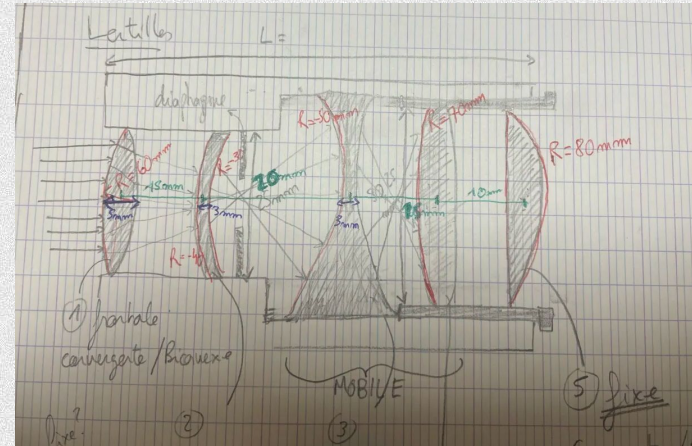
Difficulties

- Understanding of the functionment
- Displacement of the lenses
- Dimensions of the inside
- Differences between the 2 cameras



Solutions to the problems

- Simplification of the mechanism
- Reduction of the number of lenses
- Inside made so that there is no clipping
- Two different assemblies



CONCLUSION



1

2 camera lenses were designed with parameters that affect the type of camera lenses and the extension/reduction of them

2

The difficulties encountered were overcome with simplification and reverse engineering on actual camera lenses

